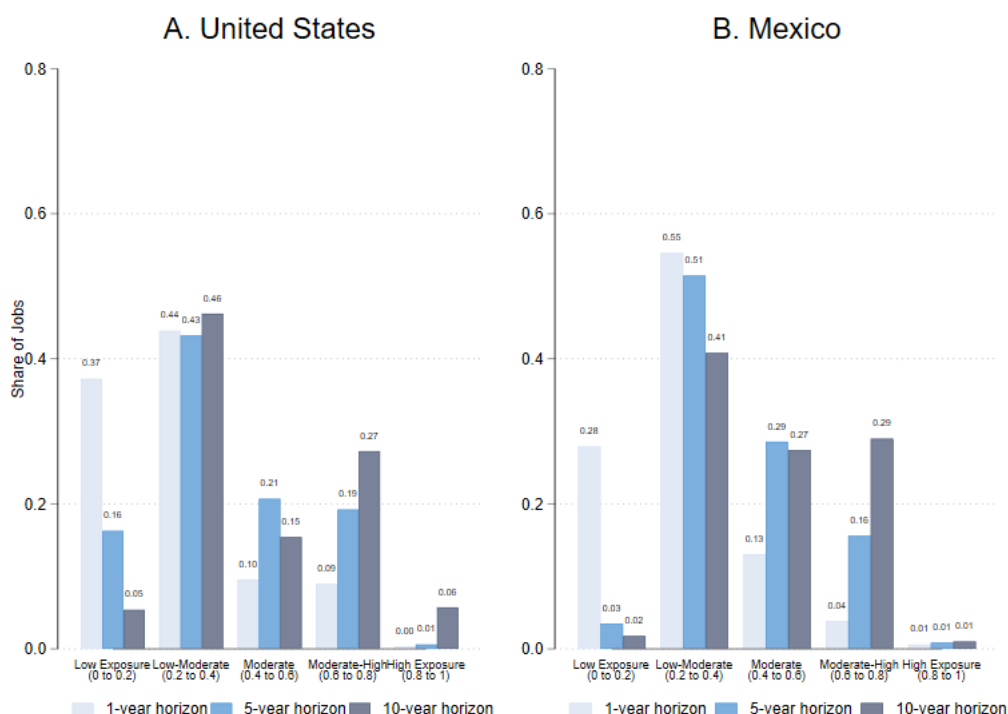


Figure 4: Distribution of Employment in the United States and Mexico by Occupational Exposure



Source: Authors' calculations using data from the Current Population Survey (CPS), Encuesta Nacional de Ocupación y Empleo (ENOE), O*NET, and surveys conducted with GPT-4 through OpenAI's API. *Note:* This figure shows the share of total jobs in each range of occupational exposure according to the GENOE index for each time horizon and country.

Table 5 presents the average exposure of the U.S. labor market to AI displacement, broken down by various socioeconomic factors. The average exposure is 0.28 for the one-year horizon, indicating that approximately 42.6 million jobs, or roughly one-quarter of total employment, are at risk of being displaced by AI adoption. Over longer time horizons, the average exposure increases to 0.39 (59.8 million jobs) and 0.46 (70.2 million jobs) for the five-year and 10-year projections, respectively.

Notably, women face higher exposure levels across all three time horizons, primarily because many work in office, administrative, and support occupations, which are among the most vulnerable to AI. Additionally, workers with lower levels of education, such as those who have not completed high school or only have a high school diploma, and individuals in the lowest earnings quintile, exhibit higher exposure scores compared to other groups. These findings suggest that AI adoption could exacerbate income inequality, echoing patterns observed in previous technological advancements (Autor et al., 2006; Autor and Dorn, 2013; Autor, 2019; Acemoglu and Restrepo, 2022).

Table 5: Occupational Exposure in the U.S. Labor Market According to GENOE

	1-year horizon	5-year horizon	10-year horizon
All workers	0.280	0.392	0.461
Sex			
Men	0.262	0.381	0.455
Women	0.298	0.404	0.466
Education level			
No High-School	0.293	0.423	0.503
High-School	0.309	0.434	0.509
Associate Degree	0.276	0.386	0.451
Bachelor's	0.261	0.357	0.417
Master's	0.199	0.281	0.333
Professional School	0.158	0.242	0.313
Doctorate	0.175	0.260	0.319
Quintiles of weekly earnings			
1st	0.339	0.469	0.549
2nd	0.335	0.464	0.538
3rd	0.320	0.446	0.518
4th	0.301	0.422	0.492
5th	0.215	0.318	0.378
Jobs exposed (in millions)	42.642	59.763	70.249

Source: Authors' calculations using data from the 2023 Current Population Survey (CPS), O*NET, and surveys conducted with GPT-4 through OpenAI's API. *Note:* This table shows the average GENOE score for each time horizon by gender, educational level, and quintiles of weekly earnings. The total jobs exposed are calculated as the average exposure multiplied by total U.S. employment in 2023.

Average occupational exposure in Mexico is quite similar to that in the United States, with 28%, 41%, and 48% of jobs exposed to AI in the one-year, five-year, and 10-year horizons, respectively (Table 6). In the immediate future, the analysis indicates that approximately 16 million jobs are at risk of being displaced by AI adoption. In contrast to the United States, the distribution of exposure by educational level and earnings is different. In Mexico, both low- and medium-skilled workers—those with post-secondary technical and professional education—face higher exposure levels. Additionally, exposure is higher for workers in the middle of the wage distribution.

This pattern probably reflects the occupational structure in Mexico (and probably of many developing economies), where office and administrative workers are more commonly found in

more middle-income positions, unlike in the United States and the developed world, where these roles are typically at the lower end of the income distribution. In Mexico, these positions are often formal jobs, more vulnerable to AI-driven automation, as also shown in the table. Conversely, informal jobs, which are typically lower on the income spectrum, show less exposure to AI. Lower exposure of informal jobs is likely due to their less intensive use of technology and computer-based tasks, as these roles tend to be more manual and less reliant on digital processes.

Table 6: Occupational Exposure in Mexico's Labor Market According to GENOE

	1-year horizon	5-year horizon	10-year horizon
All workers	0.288	0.410	0.481
Sex			
Men	0.279	0.402	0.472
Women	0.303	0.423	0.494
Type of worker			
Formal	0.306	0.422	0.487
Informal	0.274	0.401	0.476
Education level			
No High-School	0.292	0.423	0.497
High-School	0.208	0.295	0.352
Technical	0.298	0.417	0.482
Bachelor's	0.284	0.385	0.447
Master's	0.213	0.296	0.349
Doctorate	0.182	0.253	0.302
Quintiles of monthly earnings			
1st	0.278	0.409	0.484
2nd	0.301	0.432	0.508
3rd	0.300	0.426	0.497
4th	0.285	0.402	0.470
5th	0.257	0.358	0.421
Jobs exposed (in millions)	15.679	22.307	26.141

Source: Authors' calculations using data from the 2023 Encuesta Nacional de Ocupación y Empleo (ENOE), O*NET, and surveys conducted with GPT-4 through OpenAI's API. *Note:* This table shows the average GENOE score for each time horizon by gender, type of worker, educational level, and quintiles of weekly earnings. The total jobs exposed are calculated as the average exposure multiplied by the total Mexican employment in 2023.

Extrapolating these findings to Latin America and the Caribbean, we estimate that approximately 84 million jobs are exposed to AI within one year, rising to 114 million in 5 years, and 132 million in 10 years. Extending this average exposure to global labor markets suggests that approximately 980 million, 1.33 billion, and 1.54 billion jobs may be exposed to AI over the one-, five-, and 10-year horizons, respectively. However, it is important to note that this projection assumes uniform AI adoption rates and labor market structures across countries, which may not accurately reflect the global economic landscape.

5 LIMITATIONS AND CONSIDERATIONS

While our methodology offers a novel and comprehensive approach to assessing occupational exposure to AI, it is important to acknowledge its limitations. Firstly, the assessments are based on the knowledge and potential biases inherent in the GPT-4o model. Although state-of-the-art, this model may have limitations in understanding certain specialized fields or future AI developments. Additionally, predictions about future AI capabilities, especially in the five and 10-year horizons, are inherently uncertain and may not account for unforeseen breakthroughs or obstacles in AI development. Our method attempts to capture broader contextual factors, but it may not fully account for all social, economic, and regulatory factors that could influence the adoption of AI in various occupations. Lastly, while the O*NET task descriptions are detailed, they may not capture all ways in which tasks are performed in real-world settings, potentially affecting the accuracy of AI exposure estimates.

Another limitation of our index—and indeed, of all indexes that aim to capture occupational exposure to technology—is that it does not account for general equilibrium effects that can influence social preferences and the demand for certain jobs. For instance, while firefighters currently exhibit low exposure estimates to technological displacement, advances in electrical systems, domestic safety features, and industrial processes could lead to a significant reduction in the occurrence of fires. This, in turn, could decrease the overall demand for firefighters, despite their low exposure to direct technological replacement. Thus, our index may not fully capture how broader technological progress can indirectly impact the demand for certain occupations through changes in societal needs and preferences.

Despite these limitations, the GENOE index offers a unique and holistic perspective on the future of work in the age of artificial intelligence. By combining a robust data source, a novel AI-driven survey methodology, and comprehensive validation exercises, we have created a valuable tool for researchers, policymakers, and industry leaders to anticipate and prepare for the potential impacts of AI on the labor market. The index’s strengths lie in its comprehensive coverage, consistent evaluation criteria, and ability to capture more complex aspects like interactions between tasks and contextual factors. As such, while acknowledging its constraints, we believe the GENOE index provides a significant contribution to our understanding of occupational exposure to AI and can inform strategic decision-making in workforce planning and policy development.